Linear Regression $\xrightarrow[\text{known as}]{\text{Also}}$ Ordinary Least Squares.

<u>Linear Regression</u> : Linear Regression is a statistical supervised learning technique to predict the quantitative variable by forming a linear relationship with one or more independent features.

↳ Regression predict a continuous value.

<u>Ex</u>:

Time Spent (X)	Marks (Y)
5	16
3	9.8
4	13.5
11	34
...	...

The goal of the algorithm is to learn a linear model that predicts a y for an unseen x with minimum error.

$$\boxed{y = f(x)}$$

- Linear Regression
- Error Function
- Gradient Descent
- Implementation

Notation

vector

	0	1	
	Time spent x_0	No. of Assignment x_1	Marks y
0	5	3	16
1	3	4	9.8
2	4	4	13.5
⋮	11	3	34
m-1	...	...	...

mx2

m → no. of training example

n → no. of features

x → input data matrix of shape (mxn)

y → target value

theta → weights/parameters

y_hat → hypothesis (output a real number)

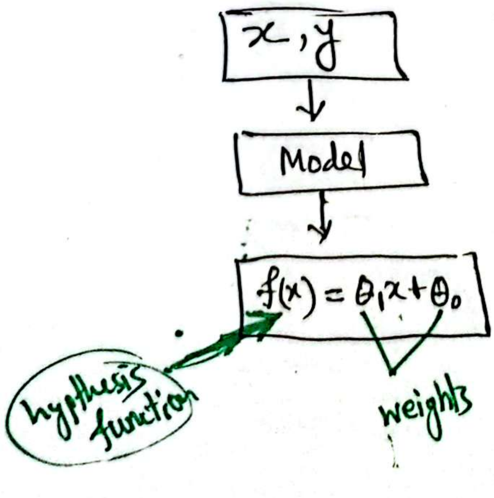
theta

$y = f(x) = mx + c$

$\Rightarrow y = \theta_1 x + \theta_0$

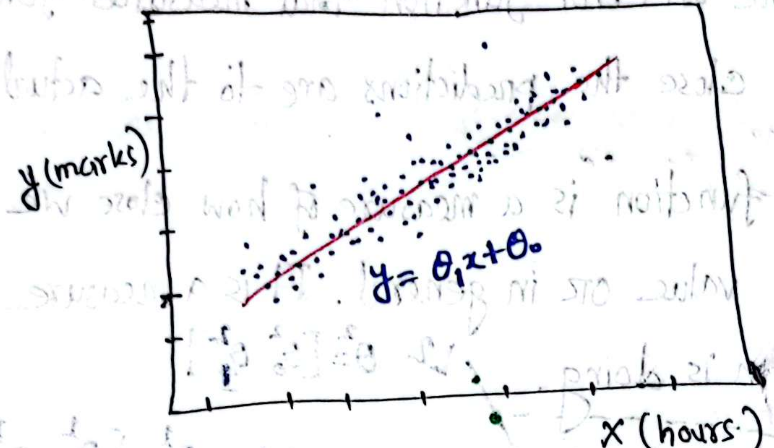
$\therefore \theta = \begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix}$

hypothesis

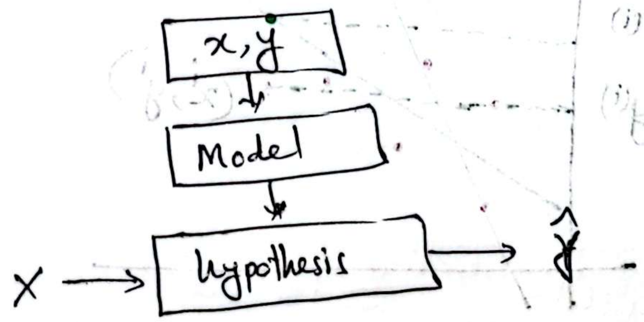


Step-1 : Assume a Hypothesis function

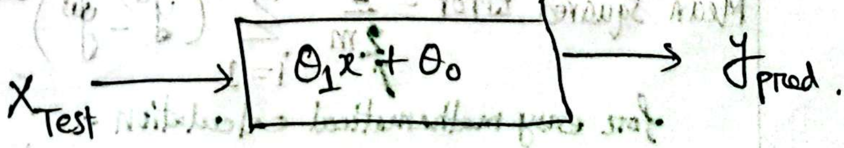
↳ Linear regression is a parametric method, which means it makes an assumption about the form of the function relating X and Y.



so we decide to approximate y as a linear function of x.



$$\boxed{h_\theta(x) = \theta_1 x + \theta_0}$$

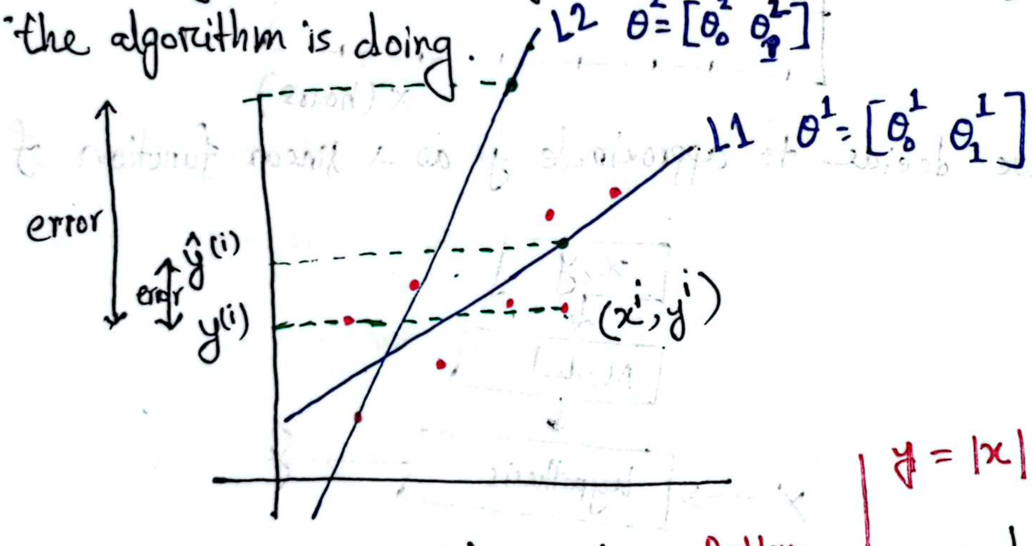


Step-2 Decide Loss/Error function

Q Given a training set, how do we learn the best values for the parameters θ?

↳ We define an error function that measures for each value of θ how close the predictions are to the actual y values.

↳ The loss function is a measure of how close we are to the true/target value or in general. It is a measure of how good the algorithm is doing.



Absulate Error $= \sum_{i=1}^{m} |\hat{y}^{(i)} - y^{(i)}|$ → Problem

Squared Error $= \sum_{i=1}^{m} (\hat{y}^{(i)} - y^{(i)})^2$

Instead of squared Error we take Mean Squared Error,

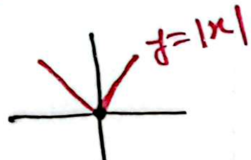
Mean Squared Error $= \frac{1}{2m} \sum_{i=1}^{m} (\hat{y}^{(i)} - y^{(i)})^2$

for easy mathematical calculation.

So, Loss Function = Mean squared Error

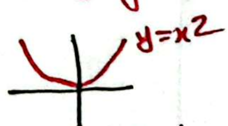
$$\boxed{J = \frac{1}{2m} \sum_{i=1}^{m} (\hat{y}^{(i)} - y^{(i)})^2}$$

$y = |x|$



x is not differentiable at $x=0$.

So, take $y = x^2$



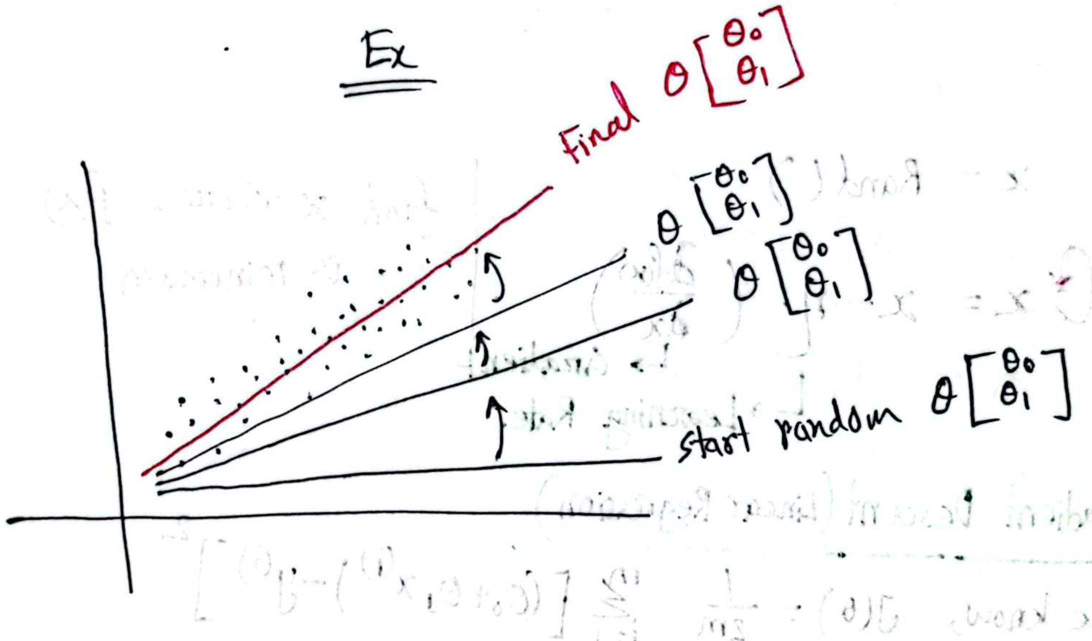
it is differentiable and doesn't need to absulate value.

Step-3 Training an algorithm that reduces error on training data

Algo: Starting with a random θ, ~~we need and~~ we need an algorithm that iteratively improves θ by reducing $J(\theta)$ errors in each step and converges eventually to minimum error.

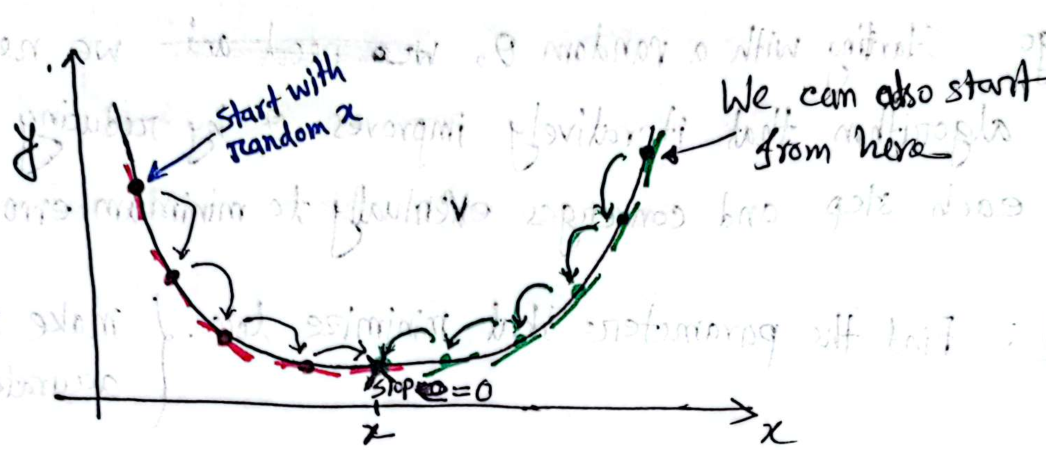
Goal: Find the parameters that minimize loss. { make model as accurate as possible

Ex



Optimization Technique

Gradient Descent Algorithm



$x = \text{Rand}()$

Loop $x = x - \eta \cdot \left(\frac{\partial f(x)}{\partial x}\right)$

η → Learning Rate

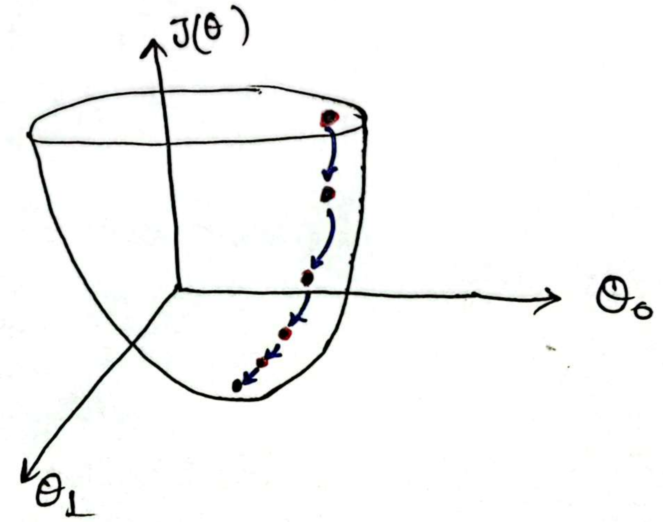
$\frac{\partial f(x)}{\partial x}$ → Gradient

find x where $f(x)$ is minimum.

Gradient Descent (Linear Regression)

We know, $J(\theta) = \frac{1}{2m} \sum_{i=1}^{m} \left[(\theta_0 + \theta_1 x^{(i)}) - y^{(i)}\right]^2$

so, $\theta = [\theta_0 \ \theta_1]$



Step-4 The Math of Training

LMS (Least Mean Squares) Update Rule

$$\boxed{\theta_j = \theta_j - \eta \cdot \frac{\partial J(\theta)}{\partial \theta_j}}$$

A partial derivative indicate how much total loss increased or decrease if we increase theta parameter by a very small ammount.

We know,

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^{m} \left[\theta_0 + \theta_1 x^{(i)} - y^{(i)}\right]^2$$

$$\therefore \frac{J(\theta)}{\partial \theta_0} = \frac{1}{2m} \cdot 2 \sum_{i=1}^{m} \left[\theta_0 + \theta_1 x^{(i)} - y^{(i)}\right] \cdot [1]$$

$$= \frac{1}{m} \sum_{i=1}^{m} \left[h_\theta(x^{(i)}) - y^{(i)}\right]$$

$$\therefore \frac{J(\theta)}{\partial \theta_1} = \frac{1}{2m} \cdot 2 \sum_{i=1}^{m} \left[\theta_0 + \theta_1 x^{(i)} - y^{(i)}\right] \left[0 + x^{(i)} + 0\right]$$

$$= \frac{1}{m} \sum_{i=1}^{m} \left[h_\theta(x^{(i)}) - y^{(i)}\right] x^{(i)}$$

Now,

Loop

$$\theta_0 = \theta_0 - \eta \cdot \frac{\partial J(\theta)}{\partial \theta_0} \quad \text{and} \quad \theta_1 = \theta_1 - \eta \cdot \frac{\partial J(\theta)}{\partial \theta_1}$$

Step-5

Coding → GitHub